Cyberster Red Team Internship

LET'S DEFEND WITH PRIDE

Weekly Report

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Field: Cybersecurity / Red Teaming

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Executive Summary

This report presents the results of reconnaissance activities conducted during **Week 01 of the Cyberstera-2026 Cybersecurity Internship Program**. The focus of this week was **Advanced Reconnaissance and Attack Surface Mapping**, aimed at identifying publicly accessible assets associated with the target domain.

The assessment targeted the training platform **Hack This Site** (hackthissite.org). The reconnaissance activities were performed using passive techniques to avoid direct interaction that could disrupt services or trigger detection mechanisms.

Multiple industry-standard security tools were used to gather information from open sources, identify subdomains, analyse infrastructure technologies, and visualize live web assets. The reconnaissance process resulted in the discovery of **hundreds of subdomains**, out of which **43 active web assets** were identified.

Additionally, technology fingerprinting revealed the use of several web technologies including an outdated version of **jQuery**, which is associated with the known vulnerability **CVE-2020-11022**.

Although no critical security vulnerabilities or credential leaks were identified during this stage, the reconnaissance process successfully mapped the organization's external attack surface and identified potential areas for deeper analysis in subsequent phases.

Weekly Objectives

The objectives for this week were:

- Understand reconnaissance methodologies used in penetration testing and red team operations
- Practice using OSINT tools for discovering organizational assets
- Identify publicly accessible infrastructure components
- Document reconnaissance findings in a structured cybersecurity report

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Task 01 — Passive OSINT & Subdomain Enumeration

Objective

To discover subdomains associated with the target domain in order to understand the organization's external attack surface.

Passive reconnaissance techniques were used to gather subdomain information from public data sources without actively interacting with the target infrastructure.

Tool: Amass

Purpose

Amass is an advanced reconnaissance tool used to collect subdomain information from numerous OSINT data sources including certificate transparency logs, DNS records, and public databases.

Command Used

amass enum -passive -d hackthissite.org | tee amass_output.txt

Result

The command returned multiple subdomains associated with the target domain.
The output was stored in amass_output.txt.

Evidence

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks]
└─$ amass enum -passive  -d hackthissite.org | tee am_output.txt
```

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks]
└─$ cat am_output.txt
hackthissite.org (FQDN) --> ns_record --> h.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> ns_record --> j.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> ns_record --> c.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> ns_record --> f.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> ns_record --> g.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> aspmx4.googlemail.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> alt2.aspmx.l.google.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> aspmx.l.google.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> alt1.aspmx.l.google.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> aspmx3.googlemail.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> aspmx5.googlemail.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> aspmx2.googlemail.com (FQDN)
irc-www.hackthissite.org (FQDN) --> cname_record --> www.irc.hackthissite.org (FQDN)
irc-lille.hackthissite.org (FQDN) --> cname_record --> lille.irc.hackthissite.org (FQDN)
0-v3dev-cdn.hackthissite.org (FQDN) --> cname_record --> v3dev.hackthissite.org (FQDN)
2-v3stage-cdn.hackthissite.org (FQDN) --> cname_record --> v3stage-cdn.hackthissite.org (FQDN
1-v3stage-cdn.hackthissite.org (FQDN) --> cname_record --> v3stage-cdn.hackthissite.org (FQDN
forums.hackthissite.org (FQDN) --> cname_record --> hackthissite.org (FQDN)
1-v3dev-cdn.hackthissite.org (FQDN) --> cname_record --> v3dev.hackthissite.org (FQDN)
4-v3stage-cdn.hackthissite.org (FQDN) --> cname_record --> v3stage-cdn.hackthissite.org (FQDN
3-v3dev-cdn.hackthissite.org (FQDN) --> cname_record --> v3dev.hackthissite.org (FQDN)
ctf.hackthissite.org (FQDN) --> cname_record --> hackthissite.github.io (FQDN)
4-v3dev-cdn.hackthissite.org (FQDN) --> cname_record --> v3dev.hackthissite.org (FQDN)
2-v3dev-cdn.hackthissite.org (FQDN) --> cname_record --> v3dev.hackthissite.org (FQDN)
h5ai.hackthissite.org (FQDN) --> cname_record --> hackthissite.github.io (FQDN)
hub.irc.hackthissite.org (FQDN) --> cname_record --> irc-hub.hackthissite.org (FQDN)
psrp.hackthissite.org (FQDN) --> cname_record --> hackmail.org (FQDN)
vm-200.outbound.firewall.hackthissite.org (FQDN) --> a_record --> 137.74.187.157 (IPAddress)
0-v3stage-cdn.hackthissite.org (FQDN) --> cname_record --> v3stage-cdn.hackthissite.org (FQDN
3-v3stage-cdn.hackthissite.org (FQDN) --> cname_record --> v3stage-cdn.hackthissite.org (FQDN
```

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks]
└─$ cat am_output.txt |wc -l
275
```

```
hackthissite.org (FQDN) --> ns_record --> h.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> ns_record --> j.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> ns_record --> c.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> ns_record --> f.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> ns_record --> g.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> aspmx4.googlemail.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> alt2.aspmx.l.google.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> aspmx.l.google.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> alt1.aspmx.l.google.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> aspmx3.googlemail.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> aspmx5.googlemail.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> aspmx2.googlemail.com (FQDN)
hackthissite.org (FQDN) --> node --> v3dev.hackthissite.org (FQDN)
hackthissite.org (FQDN) --> node --> forums.hackthissite.org (FQDN)
hackthissite.org (FQDN) --> node --> 2-v3dev-cdn.hackthissite.org (FQDN)
hackthissite.org (FQDN) --> node --> status.hackthissite.org (FQDN)
forums.hackthissite.org (FQDN) --> cname_record --> hackthissite.org (FQDN
2-v3dev-cdn.hackthissite.org (FQDN) --> cname_record --> v3dev.hackthissit
status.hackthissite.org (FQDN) --> cname_record --> edge.redirect.pizza (F
```

Figure 1 — Passive subdomain enumeration using Amass

Tool: Subfinder

Purpose

Subfinder is a passive reconnaissance tool used for fast subdomain discovery by querying multiple OSINT data providers.

Command Used

subfinder -d hackthissite.org -silent -o sf_output.txt

Result

Additional subdomains were discovered and stored in sf_output.txt.

Evidence

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks]
└─$ subfinder -d hackthissite.org -silent -o sf_output.txt
```

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks]
└─$ cat sf_output.txt
irc-v6.hackthissite.org
api.hackthissite.org
vm-099.outbound.firewall.hackthissite.org
irc-hub.hackthissite.org
pi.hackthissite.org
git.hackthissite.org
stats.hackthissite.org
wolf.irc.hackthissite.org
irc-wolf.hackthissite.org
jupiter.hackthissite.org
4-v3stage-cdn.hackthissite.org
vm-050.outbound.firewall.hackthissite.org
lille.irc.hackthissite.org
mirror.hackthissite.org
daemon.hackthissite.org
vm-005.outbound.firewall.hackthissite.org
legal.hackthissite.org
mta-sts.hackthissite.org
ns2.hackthissite.org
staff.hackthissite.org
```

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks]
└─$ cat sf_output.txt|wc -l
57
```

Figure 2 — Subdomain discovery using Subfinder

Tool: Assetfinder

Purpose

Assetfinder identifies related domains and subdomains associated with an organization by querying public sources.

Command Used

assetfinder --subs-only hackthissite.org > af_output.txt

Result

The command produced additional subdomain results which were saved to assetfinder_output.txt.

Evidence

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks]
└─$ assetfinder -subs-only hackthissite.org > af_output.txt
```

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks]
└─$ cat af_output.txt
hackthissite.org
api.hackthissite.org
ctf.hackthissite.org
email.hackthissite.org
forums.hackthissite.org
git.hackthissite.org
h5ai.hackthissite.org
hp.hackthissite.org
irc.hackthissite.org
irc-hub.hackthissite.org
irc-www.hackthissite.org
lille.irc.hackthissite.org
wolf.irc.hackthissite.org
www.irc.hackthissite.org
legal.hackthissite.org
mirror.hackthissite.org
psrp.hackthissite.org
qdb.hackthissite.org
stats.hackthissite.org
status.hackthissite.org
www.hackthissite.org
```

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks]
└─$ cat af_output.txt | wc -l
21
```

Figure 3 — Subdomain enumeration using Assetfinder

Merging Enumeration Results

To create a clean dataset, outputs from the enumeration tools were combined and duplicates were removed.

Command Used

cat *.txt | sort -u > final_subdomains.txt

Result

Metric	Result
Total Subdomains Collected	411
Unique Subdomains	217

Tool: DNSx

Purpose

DNSx verifies whether discovered subdomains resolve to valid IP addresses, helping identify active assets.

Command Used

dnsx -l final_subdomains.txt -o resolved.txt

Result

Metric	Result
Resolving Subdomains	43

Evidence

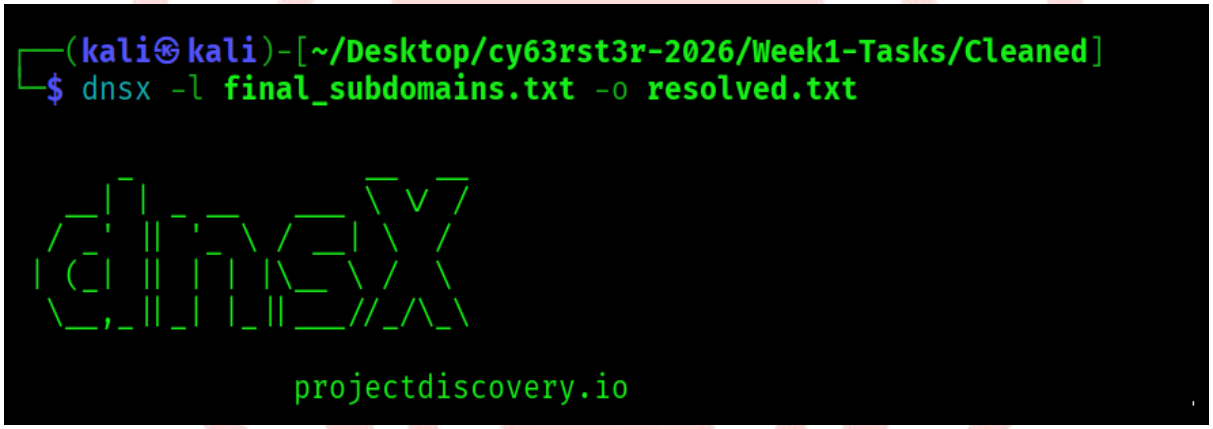

Figure 4 — DNS resolution verification using DNSx

Task 02 — Infrastructure & Technology Profiling

Objective

To identify the technologies used by the target web infrastructure, including web servers, frameworks, and programming languages.

Understanding the technology stack helps security professionals identify potential vulnerabilities related to outdated software or misconfigurations.

Tool: WhatWeb

Purpose

WhatWeb is used for technology fingerprinting, allowing analysts to identify web technologies used by a website.

Command Used

Whatweb -v https://hackthissite.org

Result

The analysis detected several technologies used by the web application.

Technology	**Detected**
Web Server	Apache
Backend Language	PH
JavaScript Library	jQuery 1.8.1
Language	PHP

Security Observation

The website was found to use **jQuery version 1.8.1**, which is outdated.

This version is associated with the vulnerability **CVE-2020-11022**, which may allow Cross-Site Scripting (XSS) attacks under certain conditions.

Evidence

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks]
└─$ whatweb -v https://hackthissite.org > ww_output.txt
```

Figure 5 — Technology fingerprinting using WhatWeb

Tool: Wappalyzer

Purpose

Wappalyzer identifies web technologies directly through browser analysis.

Method Used

The Wappalyzer browser extension was used to inspect the technologies running on the website.

Result

The extension confirmed the presence of:

- Apache Web Server
-
- PHP Backend
- jQuery JavaScript library

Evidence

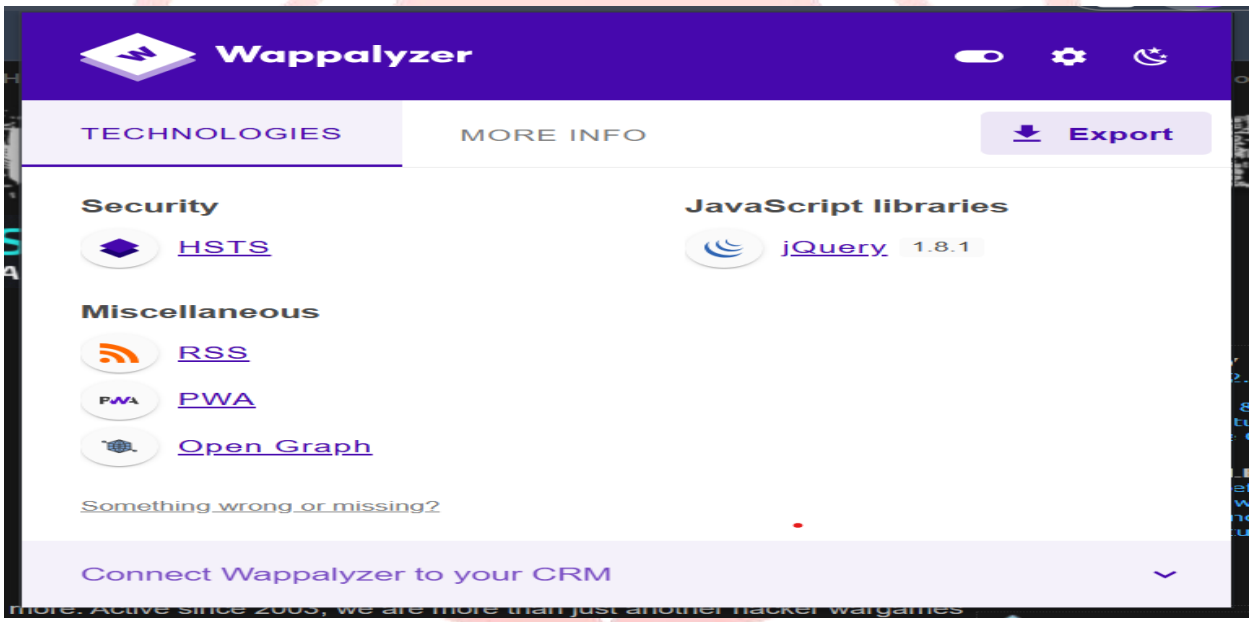

Figure 6 — Technology detection using Wappalyzer

Task 03 — GitHub Dorking & Credential Leak Detection

Objective

To identify potential credential leaks or sensitive information exposed in public repositories.

Tool: GitHacker

Purpose

GitHacker is used to retrieve exposed Git repositories and analyse commit history for sensitive information.

Command Used

Python3 -m githacker venv

Source githacker/bin/activate

Githacker –url https://hackthissite.org –output-folder ./Githacker_Report

Result

No exposed repositories or credentials related to the target organization were discovered during this analysis.

Evidence

```
┌──(githacker)─(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks/Cleaned]
└─$ githacker --url https://hackthissite.org/.git --output-folder ./GitHacker_Report
2026-03-06 15:53:06 INFO 1 urls to be exploited
2026-03-06 15:53:06 INFO Exploiting https://hackthissite.org/.git into ./GitHacker_Report/2b8
/home/kali/githacker/lib/python3.13/site-packages/urllib3/connectionpool.py:1097: InsecureReq
to host 'hackthissite.org'. Adding certificate verification is strongly advised. See: https:/
l#tls-warnings
  warnings.warn(
/home/kali/githacker/lib/python3.13/site-packages/urllib3/connectionpool.py:1097: InsecureReq
to host 'www.hackthissite.org'. Adding certificate verification is strongly advised. See: htt
.html#tls-warnings
  warnings.warn(
/home/kali/githacker/lib/python3.13/site-packages/urllib3/connectionpool.py:1097: InsecureReq
to host 'hackthissite.org'. Adding certificate verification is strongly advised. See: https:/
l#tls-warnings
  warnings.warn(
/home/kali/githacker/lib/python3.13/site-packages/urllib3/connectionpool.py:1097: InsecureReq
to host 'www.hackthissite.org'. Adding certificate verification is strongly advised. See: htt
.html#tls-warnings
  warnings.warn(
/home/kali/githacker/lib/python3.13/site-packages/urllib3/connectionpool.py:1097: InsecureReq
to host 'hackthissite.org'. Adding certificate verification is strongly advised. See: https:/
l#tls-warnings
  warnings.warn(
2026-03-06 15:53:25 ERROR The target url(https://hackthissite.org/) is not a valid git reposi
2026-03-06 15:53:25 INFO 0 / 1 were exploited successfully
```

Figure 7 — GitHacker repository scan

Manual GitHub Dorking

Platform used: GitHub

Purpose

Manual GitHub search queries were used to identify potential credential leaks.

Queries Used

hackthissite API_KEY

hackthissite PASSWORD

hackthissite AWS_SECRET

hackthissite SQL_CONNECTION

Result

No sensitive credentials were discovered for hackthissite.org during manual searches but exposed other targets out of scope.

However, this process demonstrates how developers may accidentally expose sensitive information in public repositories.

Evidence

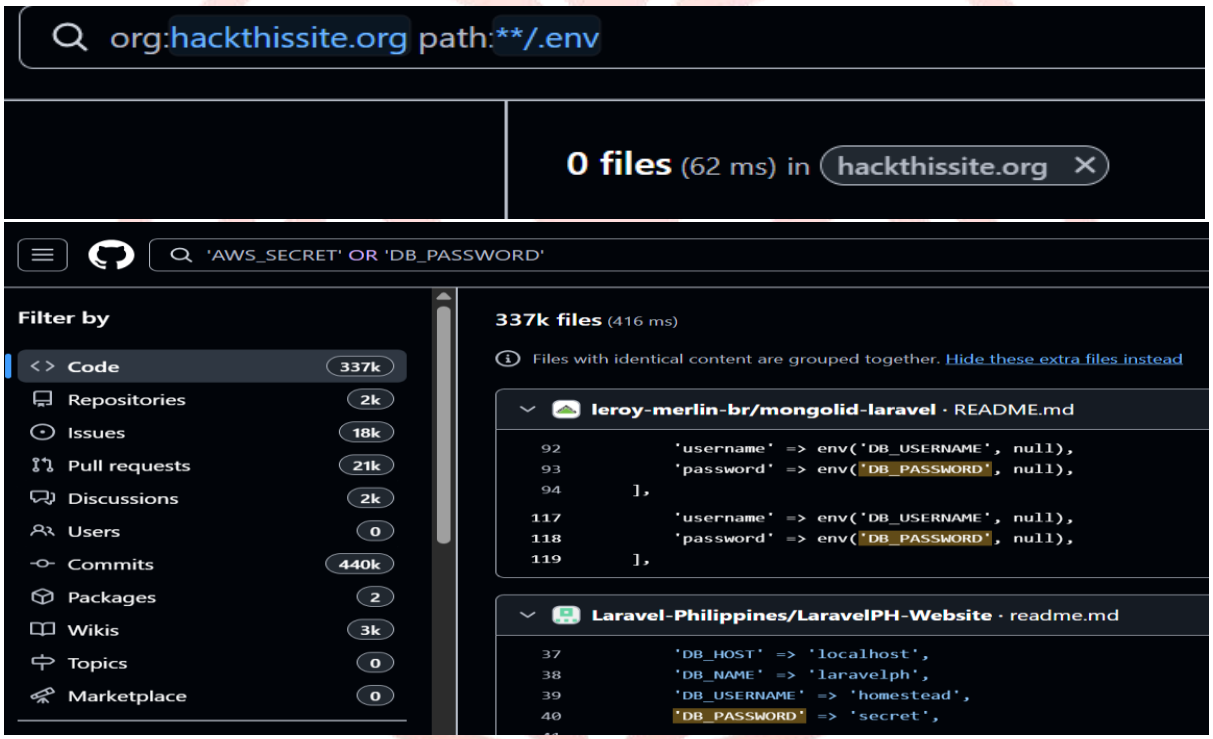

Figure 8 — Manual GitHub dorking

Task 04 — Visual Reconnaissance & Live Asset Identification

Objective

To identify which discovered subdomains are actively hosting web services and capture screenshots for visual reconnaissance.

Tool: httprobe and Httpx

Purpose

Httprobe verifies which subdomains have active HTTP or HTTPS services.

Command Used

cat sub-resolved.txt | httprobe > live.txt

Result

Metric	Result
Live Web Assets	44

Evidence

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks/Cleaned]
└─$ cat sub-resolved.txt | httprobe > live.txt
```

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks/Cleaned]
└─$ cat live.txt |wc -l
44
```

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks/Cleaned]
└─$ cat live.txt
http://0-v3dev-cdn.hackthissite.org
http://0-v3stage-cdn.hackthissite.org
http://1-v3dev-cdn.hackthissite.org
http://1-v3stage-cdn.hackthissite.org
http://2-v3dev-cdn.hackthissite.org
http://2-v3stage-cdn.hackthissite.org
http://3-v3dev-cdn.hackthissite.org
http://3-v3stage-cdn.hackthissite.org
http://4-v3dev-cdn.hackthissite.org
http://4-v3stage-cdn.hackthissite.org
http://ctf.hackthissite.org
http://forums.hackthissite.org
http://h5ai.hackthissite.org
http://hackthissite.org
http://hp.hackthissite.org
http://hub.irc.hackthissite.org
http://irc.hackthissite.org
http://irc-hub.hackthissite.org
http://irc-lille.hackthissite.org
http://irc-wolf.hackthissite.org
http://irc-www.hackthissite.org
http://lille.irc.hackthissite.org
http://mirror.hackthissite.org
http://psrp.hackthissite.org
```

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks/Cleaned]
└─$ ~/go/bin/httpx -l sub-resolved.txt -o live1.txt


    projectdiscovery.io

[INF] Current httpx version v1.3.5 (outdated)
https://wolf.irc.hackthissite.org
https://ctf.hackthissite.org
https://irc.hackthissite.org
https://hackthissite.org
https://h5ai.hackthissite.org
https://stats.hackthissite.org
https://status.hackthissite.org
https://legal.hackthissite.org
https://mirror.hackthissite.org
https://irc-lille.hackthissite.org
https://www.irc.hackthissite.org
http://psrp.hackthissite.org
https://www.hackthissite.org
https://irc-wolf.hackthissite.org
https://irc-www.hackthissite.org
https://hp.hackthissite.org
https://qdb.hackthissite.org
https://forums.hackthissite.org
http://1-v3dev-cdn.hackthissite.org
http://v3dev.hackthissite.org
http://1-v3stage-cdn.hackthissite.org
http://3-v3dev-cdn.hackthissite.org
http://v3stage.hackthissite.org
http://4-v3stage-cdn.hackthissite.org
http://v3stage-cdn.hackthissite.org
http://lille.irc.hackthissite.org
http://4-v3dev-cdn.hackthissite.org
http://3-v3stage-cdn.hackthissite.org
http://irc-hub.hackthissite.org
http://0-v3stage-cdn.hackthissite.org

┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks/Cleaned]
└─$ cat live1.txt | wc -l
35
```

Figure 9 — Live host discovery using httprobe

Tool: Gowitness

Purpose

Gowitness captures screenshots of active web applications to assist in visual reconnaissance.

Command Used

gowitness scan file -f live.txt –write-db –screenshot-path Gowitness_Report

Result

Screenshots of all live web assets were captured and organized into a report.

Evidence

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks/Cleaned]
└─$ gowitness scan file -f live.txt --write-db --screenshot-path Gowitness_Report
```

```
┌──(kali㉿kali)-[~/…/cy63rst3r-2026/Week1-Tasks/Cleaned/Gowitness_Report]
└─$ ls
http---ctf.hackthissite.org-443.jpeg
http---ctf.hackthissite.org-80.jpeg
http---forums.hackthissite.org-443.jpeg
http---forums.hackthissite.org-80.jpeg
http---h5ai.hackthissite.org-443.jpeg
http---h5ai.hackthissite.org-80.jpeg
http---hackthissite.org-443.jpeg
http---hackthissite.org-80.jpeg
http---hp.hackthissite.org-443.jpeg
http---hp.hackthissite.org-80.jpeg
http---irc.hackthissite.org-443.jpeg
http---irc.hackthissite.org-80.jpeg
http---irc-lille.hackthissite.org-443.jpeg
http---irc-lille.hackthissite.org-80.jpeg
```

Figure 10 — Visual reconnaissance using Gowitness

Tool: Aquatone

Purpose

Aquatone automates web asset discovery and screenshot generation.

Command Used

cat live.txt | aquatone -out Aquatone_report

Result

A structured report containing screenshots and HTTP responses of all live assets was generated.

Evidence

```
┌──(kali㉿kali)-[~/Desktop/cy63rst3r-2026/Week1-Tasks/Cleaned]
└─$ cat live.txt | aquatone -out Aquatone_Report
aquatone v1.7.0 started at 2026-03-06T08:11:41-05:00

Targets     : 44
Threads     : 2
Ports       : 80, 443, 8000, 8080, 8443
Output dir  : Aquatone_Report
```

```
Generating HTML report ... done

Writing session file ... Time:
 - Started at   : 2026-03-06T08:11:41-05:00
 - Finished at  : 2026-03-06T08:20:50-05:00
 - Duration     : 9m9s

Requests:
 - Successful : 35
 - Failed     : 9

 - 2xx : 20
 - 3xx : 0
 - 4xx : 15
 - 5xx : 0

Screenshots:
 - Successful : 0
 - Failed     : 35

Wrote HTML report to: Aquatone_Report/aquatone_report.html
```

Figure 11 — Aquatone visual asset report

Initial Security Observations

During reconnaissance, the robots.txt file revealed hidden paths:

User-agent: *

Disallow: /missions/

Disallow: /killing/all/humans/

These directories may contain non-indexed content and represent potential areas for further investigation.

Conclusion

The reconnaissance activities conducted during Week 01 successfully identified the external attack surface of the target domain. By leveraging multiple OSINT tools and techniques, numerous subdomains and live web assets associated with the organization were discovered and analysed.

Technology profiling revealed the use of several web technologies including an outdated version of jQuery with a known CVEs such as CVE-2015-9251, CVE-2019-11358, CVE-2020-11022, CVE-2020-11023 making it a LOW HANGING FRUIT, which may introduce potential vulnerabilities if not properly mitigated.

Although no critical security vulnerabilities were discovered during this phase, the findings provide a strong foundation for deeper vulnerability analysis and exploitation activities in future assessments.